

When Do Humans and Large Language Models Disagree About Chess Difficulty?

Research Question, Contribution, and Project Design

Large language models (LLMs) have shown strong performance across many reasoning tasks, but their abilities can become less reliable when problems require precise, multi-step decision making. Chess provides a useful environment for studying these limitations because every position has clearly defined legal moves and can be evaluated objectively, while the difficulty of a position can also vary substantially for human players. I am proposing a study on whether chess puzzles that are difficult for humans are also difficult for LLMs, and what features of a chess position explain the differences. I plan to use approximately 10,000+ puzzles from the Lichess open puzzle database, where puzzle ratings are derived from human solving performance through a Glicko-2 rating system and can therefore serve as a proxy for human difficulty. Prior research has already shown that LLM performance tends to decline as chess difficulty increases, but work such as Easy2Hard-Bench focuses primarily on performance across difficulty levels and excludes multi-step puzzles, leaving room to examine more complex positions and the specific features associated with LLM failure (Ding et al., 2024). I hope to contribute by identifying whether characteristics such as tactical theme, solution length, number of legal moves, game phase, number of pieces, and the type of best move help explain why some puzzles are disproportionately difficult for LLMs compared with humans. I will have 2-3 LLMs attempt the same sampled puzzles under standardized conditions and use methods such as descriptive analysis, logistic regression, and a tree-based machine learning model to measure how human difficulty and position characteristics relate to the probability of solving a puzzle correctly. The expected output will be a research paper accompanied by reproducible Python code and a derived dataset containing puzzle characteristics, LLM responses, and model performance results. The significance of this study lies in moving beyond overall LLM chess performance to examine where human and machine perceptions of difficulty diverge, which could also

help researchers develop AI reasoning benchmarks that reveal specific weaknesses rather than relying only on overall accuracy.

Preparation, Skills, and Methods

Through both coursework and previous internships, I have accumulated significant experience with Python, data cleaning, statistical modeling, machine learning, and evaluating large language models. In particular, my summer experience in building and evaluating LLM classification systems has given me practice designing prompts, measuring model accuracy, processing model outputs, and comparing model performance across many observations, all of which directly apply to this study. I have also worked with methods such as logistic regression and classification through DA 352, which can be used to estimate how factors such as puzzle rating, tactical theme, solution length, and board complexity affect the probability that an LLM solves a puzzle correctly. The main skills I will need to develop are more specific to chess analysis, including parsing FEN board positions, determining legal moves, extracting structural features from a chessboard, and using Stockfish to evaluate move quality when necessary. To develop these skills, I plan to use the python-chess library and its documentation, Stockfish documentation, examples from existing chess-LLM research when needed. I will also use Github as my main version control system to track changes throughout this study.

Scholarly Background and Research Gap

Research on both human chess expertise and artificial intelligence suggests that chess difficulty is determined by more than simply finding an objectively strong move. Recent work demonstrates that machine and human chess behavior can differ substantially: traditional engines that play at superhuman levels do not necessarily predict human decisions well, whereas human-trained systems such as Maia more accurately reproduce decisions made by players at different skill levels (McIlroy-Young et al., 2020). Studies on large language models further shows that chess remains a challenging reasoning domain, with model performance varying across structural understanding, tactical patterns, calculation,

position evaluation, and full-game strategic reasoning (Wang et al., 2025; Wen et al., 2025; Liu et al., 2026). Most directly related to my project, Easy2Hard-Bench uses human Lichess performance and Glicko-2 ratings to construct continuous chess-puzzle difficulty measures and finds that LLM performance generally changes as human-derived difficulty increases (Ding et al., 2024). However, that benchmark excludes puzzles requiring multi-step solutions and primarily examines performance across difficulty levels rather than explaining why particular positions are disproportionately difficult for language models. Other recent benchmarks similarly emphasize aggregate chess ability or individual reasoning categories rather than directly modeling the gap between human-derived difficulty and LLM performance at the position level (Wen et al., 2025; Kolasani et al., 2025). My project therefore builds on this literature by treating human-LLM difficulty disagreement as the primary outcome and testing whether interpretable features such as tactical theme, solution length, number of legal moves, game phase, and board complexity explain where human and LLM difficulty patterns diverge.

Project Timeline and Work Plan

To ensure the project can be completed within the semester, I will divide the work into clear stages that move from research design and data preparation to model testing, analysis, and final writing. The timeline below also includes a pilot stage before implementing the full LLM pipeline so that I can adjust the sample size, prompts, or evaluation process if needed.

Week Number	Overview	Description
Week 3	Finalize topic and proposal	Finalize the research question, research gap, variables, expected methods, and project scope.
Weeks 4-5	Literature review and data setup	Complete the main literature review, download the Lichess puzzle dataset, and identify the variables and puzzle characteristics needed for the project.
Weeks 6-7	Build analysis pipeline	Clean the data and use Python and python-chess to extract features such as legal moves, number of pieces, material balance, game phase, and solution length.
Weeks 8	Select puzzle sample	Create a balanced sample of about 10,000+ puzzles across different ratings, themes, and solution lengths.
Weeks 9	Pilot LLM testing	Test 50-100 puzzles on the selected LLMs, finalize prompts and scoring rules, and check runtime, cost, and response quality.
Weeks 10-11	Full data collection	Run the full puzzle sample through the selected LLMs and combine model responses with Lichess ratings and engineered chess features.
Weeks 12-13	Analyze results	Perform exploratory analysis, logistic regression, and a tree-based model to identify which characteristics predict LLM success and human-LLM difficulty differences.
Weeks 14-15	Final paper and presentation	Interpret results, create figures and tables, complete the paper, clean the code and dataset, and prepare the final presentation.

Works Cited

Ding, Mucong, et al. “Easy2Hard-Bench: Standardized Difficulty Labels for Profiling LLM Performance and Generalization.” *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, 2024. <https://doi.org/10.52202/079017-1407>.

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